Advanced package aspects

Licensing report

Licensing report: introduction

- A key aspect of embedded Linux systems is **license compliance**.
- Embedded Linux systems integrate together a number of open-source components, each distributed under its own license.
- The different open-source licenses may have **different requirements**, that must be met before the product using the embedded Linux system starts shipping.
- Buildroot helps in this license compliance process by offering the possibility of generating a number of **license-related information** from the list of selected packages.
- Generated using:

```
$ make legal-info
```

Licensing report: contents of `legal-info`

- `sources/` and `host-sources/`, all the source files that are redistributable (tarballs, patches, etc.)
- `manifest.csv` and `host-manifest.csv`, CSV files with the list of *target* and *host* packages, their version, license, etc.
- `licenses/` and `host-licenses/<pkg>/`, the full license text of all *target* and *host* packages, per package
- `buildroot.config`, the Buildroot `.config` file
- `legal-info.sha256` hashes of all *legal-info* files
- `README`

Including licensing information in packages

- `<pkg>_LICENSE`
  - Comma-separated **list of license(s)** under which the package is distributed.
  - Must use SPDX license codes, see https://spdx.org/licenses/
  - Can indicate which part is under which license (programs, tests, libraries, etc.)
- `<pkg>_LICENSE_FILES`
  - Space-separated **list of file paths** from the package source code containing the license text and copyright information
  - Paths relative to the package top-level source directory
- `<pkg>_REDISTRIBUTE`
  - Boolean indicating whether the package source code can be redistributed or not (part of the `legal-info` output)
  - Defaults to `YES`, can be overridden to `NO`
  - If `NO`, source code is not copied when generating the licensing report

Licensing information examples

linux.mk

```
LINUX_LICENSE = GPL-2.0
LINUX_LICENSE_FILES = COPYING
```

acl.mk

```
ACL_LICENSE = GPL-2.0+ (programs), LGPL-2.1+ (libraries)
ACL_LICENSE_FILES = doc/COPYING doc/COPYING.LGPL
```

owl-linux.mk

```
OWL_LINUX_LICENSE = PROPRIETARY
OWL_LINUX_LICENSE_FILES = LICENSE
OWL_LINUX_REDISTRIBUTE = NO
```

Security vulnerability tracking

Security vulnerability tracking

- Security has obviously become a key issue in embedded systems that are more and more commonly connected.
- Embedded Linux systems typically integrate 10-100+ open-source components $arrow.r$ not easy to keep track of their potential security vulnerabilities
- Industry relies on *Common Vulnerability Exposure* (CVE) reports to document known security issues
- Buildroot is able to identify if packages are affected by known CVEs, by using the *National Vulnerability Database*
  - `make pkg-stats`
  - Produces `$(O)/pkg-stats.html`, `$(O)/pkg-stats.json`
- Note: this is limited to known CVEs. It does not guarantee the absence of security vulnerabilities.
- Only applies to open-source packages, not to your own custom code.

Example pkg-stats output

package/libao/libao.mk	0	**autotools** target	Yes	Yes	Yes	1.2.0	**1.2.0** found by distro	0	Link	CVE-2017-11548	cpe:2.3:a:xiph:libao:1.2.0:*:*:*:*:*:*:*
package/libcue/libcue.mk	0	**cmake** target	Yes	Yes	Yes	2.2.1	**2.2.1** found by distro	0	Link	N/A	no verified CPE identifier
package/libebur128/libebur128.mk	0	**cmake** target	Yes	Yes	Yes	1.2.4	**1.2.6** found by distro	0	Link	N/A	no verified CPE identifier
package/libffi/libffi.mk	7	**autotools** target + host	Yes	Yes	Yes	3.3	**3.3** found by distro	0	Link	N/A	cpe:2.3:a:libffi_project:libffi:3.3:rc0:*:*:*:*:*:*
package/libglib2/libglib2.mk	4	**meson** target + host	Yes	Yes	Yes	2.66.7	**2.68.1** found by distro	0	Link	CVE-2021-28153	cpe:2.3:a:gnome:glib:2.66.7:*:*:*:*:*:*:*
package/libid3tag/libid3tag.mk	0	**autotools** target	Yes	Yes	Yes	0.15.1b	**0.15.1b** found by distro	0	Link	N/A	no verified CPE identifier
package/liblo/liblo.mk	0	**autotools** target	Yes	Yes	Yes	0.31	**0.31** found by distro	0	Link	N/A	no verified CPE identifier
package/libmad/libmad.mk	2	**autotools** target	Yes	Yes	Yes	0.15.1b	**0.15.1b** found by distro	0	Link	CVE-2018-7263	no verified CPE identifier
package/libmodplug/libmodplug.mk	0	**autotools** target	Yes	Yes	Yes	0.8.9.0	**0.8.9.0** found by distro	0	Link	N/A	cpe:2.3:a:konstanty_bialkowski:libmodplug:0.8.9.0:*:*:*:*:*:*:* CPE identifier unknown in CPE database
package/libmpd/libmpd.mk	1	**autotools** target	Yes	Yes	Yes	11.8.17	**11.8.17** found by distro	0	invalid 502	N/A	no verified CPE identifier
package/libtool/libtool.mk	0	**autotools** target + host	Yes	Yes	Yes	2.4.6	**2.4.6** found by distro	0	Link	N/A	no verified CPE identifier

Packages affected by CVEs	5
Total number of CVEs affecting all packages	5
Packages with CPE ID	13
Packages without CPE ID	30

CPE: Common Platform Enumeration

- Concept of *Common Platform Enumeration*, which gives a unique identifier to a software release
  - E.g.: `cpe:2.3:a:xiph:libao:1.2.0:*:*:*:*:*:*:*`
- By default Buildroot uses:
  - `cpe:2.3:a:<pkg>_project:<pkg>:<pkg>_VERSION:*:*:*:*:*:*:*`
  - Not always correct!
- Can be modified using:
  - `<pkg>_CPE_ID_PREFIX`
  - `<pkg>_CPE_ID_VENDOR`
  - `<pkg>_CPE_ID_PRODUCT`
  - `<pkg>_CPE_ID_VERSION`
  - `<pkg>_CPE_ID_UPDATE`
- Concept of *CPE dictionary* provided by NVD, which contains all known CPEs.
  - `pkg-stats` checks if the CPE of each package is known in the *CPE dictionary*

NVD CVE-2020-35492 example

https://nvd.nist.gov/vuln/detail/CVE-2020-35492

Known Affected Software Configurations Switch to CPE 2.2

Configuration 1 (hide)

cpe:2.3:a:cairographics:cairo:*:*:*:*:*:*:*:*	Up to (excluding)
Hide Matching CPE(s)	1.17.4

Hide Matching CPE(s) ▲

- *cpe:2.3:a:cairographics:cairo:-:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.0.0:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.0.2:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.0.4:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.2.0:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.2.2:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.2.4:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.2.6:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.4.0:*:*:*:*:*:*:**
- *cpe:2.3:a:cairographics:cairo:1.4.2:*:*:*:*:*:*:**

Showing 10 of 50 matching CPE(s) for the range. View All CPEs here

CPE information in packages

package/bash/bash.mk

```
BASH_CPE_ID_VENDOR = gnu
```

package/audit/audit.mk

```
AUDIT_CPE_ID_VENDOR = linux_audit_project AUDIT_CPE_ID_PRODUCT = linux_audit
```

linux/linux.mk

```
LINUX_CPE_ID_VENDOR = linux LINUX_CPE_ID_PRODUCT = linux_kernel
LINUX_CPE_ID_PREFIX = cpe:2.3:o
```

package/libffi/libffi.mk

```
LIBFFI_CPE_ID_VERSION = 3.3
LIBFFI_CPE_ID_UPDATE = rc0
```

<pkg>_IGNORE_CVES variable

- There are cases where a CVE reported by the *pkg-stats* tool in fact is not relevant:
  - The security fix has been backported into Buildroot
  - The vulnerability does not affect Buildroot due to how the package is configured or used
- The `<pkg>_IGNORE_CVES` variable allows a package to tell *pkg-stats* to ignore a particular CVE

package/bind/bind.mk

```
#Only applies to RHEL6.x with DNSSEC validation on BIND_IGNORE_CVES =
CVE-2017-3139
```

package/avahi/avahi.mk

```
#0001-Fix-NULL-pointer-crashes-from-175.patch AVAHI_IGNORE_CVES +=
CVE-2021-36217
```

CycloneDX SBOM

- Buildroot can generate a SBOM (Software Bill Of Material) matching the standard `CycloneDX` format

```
$ make show-info | ./utils/generate-cyclonedx -o buildroot.sbom
```

- This can be used to document the contents of the build
- But also for vulnerability tracking, for example in conjunction with tools such as `Dependency Track`

Patching packages

Patching packages: why?

- In some situations, it might be needed to patch the source code of certain packages built by Buildroot.
- Useful to:
  - Fix cross-compilation issues
  - Backport bug or security fixes from upstream
  - Integrate new features or fixes not available upstream, or that are too specific to the product being made
- Patches are automatically applied by Buildroot, during the *patch* step, i.e. after extracting the package, but before configuring it.
- Buildroot already comes with a number of patches for various packages, but you may need to add more for your own packages, or to existing packages.

Patch application ordering

- Overall the patches are applied in this order:
  1. Patches mentioned in the `<pkg>_PATCH` variable of the package `.mk` file. They are automatically downloaded before being applied.
  2. Patches present in the package directory `package/<pkg>/*.patch`
  3. Patches present in the *global patch directories*
- In each case, they are applied:
  - In the order specified in a `series` file, if available
  - Otherwise, in alphabetic ordering

Patch conventions

- There are a few conventions and best practices that the Buildroot project encourages to use when managing patches
- Their name should start with a sequence number that indicates the ordering in which they should be applied.

ls package/nginx/*.patch

```
0001-auto-type-sizeof-rework-autotest-to-be-cross-compila.patch
0002-auto-feature-add-mechanism-allowing-to-force-feature.patch
0003-auto-set-ngx_feature_run_force_result-for-each-featu.patch
0004-auto-lib-libxslt-conf-allow-to-override-ngx_feature_.patch
0005-auto-unix-make-sys_nerr-guessing-cross-friendly.patch
[...]
```

- Each patch should contain a description of what the patch does, and if possible its upstream status.
- Each patch should contain a `Signed-off-by` that identifies the author of the patch.
- Patches should be generated using `git format-patch` when possible.

Patch example

```
From 81289d1d1adaf5a767a4b4d1309c286468cfd37f Mon Sep 17 00:00:00 2001
From: Samuel Martin <s.martin49@gmail.com>
Date: Thu, 24 Apr 2014 23:27:32 +0200
Subject: [PATCH] auto/type/sizeof: rework autotest to be cross-compilation
 friendly

Rework the sizeof test to do the checks at compile time instead of at runtime. This way, it does not break when cross-
compiling for a different CPU architecture.

Signed-off-by: Samuel Martin <s.martin49@gmail.com>
---
 auto/types/sizeof | 42 ++++++++++++++++++++++++++++--------------
 1 file changed, 28 insertions(+), 14 deletions(-)

diff --git a/auto/types/sizeof b/auto/types/sizeof index 9215a54..c2c3ede 100644
--- a/auto/types/sizeof
+++ b/auto/types/sizeof
@@ -14,7 +14,7 @@ END

 ngx_size=

-cat << END > \$NGX_AUTOTEST.c
+cat << \_EOF > \$NGX_AUTOTEST.c
[...]
```

Global patch directories

- You can include patches for the different packages in their package directory, `package/<pkg>/`.
- However, doing this involves changing the Buildroot sources themselves, which may not be appropriate for some highly specific patches.
- The *global patch directories* mechanism allows to specify additional locations where Buildroot will look for patches to apply on packages.
- `BR2_GLOBAL_PATCH_DIR` specifies a space-separated list of directories containing patches.
- These directories must contain sub-directories named after the packages, themselves containing the patches to be applied.

Global patch directory example

Patching *strace*

```
$ ls package/strace/*.patch
0001-linux-aarch64-add-missing-header.patch

$ find ~/patches/
~/patches/
~/patches/strace/
~/patches/strace/0001-Demo-strace-change.patch

$ grep ^BR2_GLOBAL_PATCH_DIR .config BR2_GLOBAL_PATCH_DIR="$(HOME)/patches"

$ make strace
[...]
>>> strace 4.10 Patching

Applying 0001-linux-aarch64-add-missing-header.patch using patch:
patching file linux/aarch64/arch_regs.h

Applying 0001-Demo-strace-change.patch using patch:
patching file README
[...]
```

Generating patches

- To generate the patches against a given package source code, there are typically two possibilities.
- Use the upstream version control system, often *Git*
- Use a tool called `quilt`
  - Useful when there is no version control system provided by the upstream project
  - https://savannah.nongnu.org/projects/quilt

Generating patches: with Git

Needs to be done outside of Buildroot: you cannot use the Buildroot package build directory.

1. Clone the upstream Git repository `git clone https://...`
2. Create a branch starting on the tag marking the stable release of the software as packaged in Buildroot `git checkout -b buildroot-changes v3.2`
3. Import existing Buildroot patches (if any) `git am /path/to/buildroot/package/<foo>/*.patch`
4. Make your changes and commit them `git commit -s -m "this is a change"`
5. Generate the patches `git format-patch v3.2`

Generating patches: with Quilt

1. Extract the package source code:
   `tar xf /path/to/dl/<foo>-<version>.tar.gz`
2. Inside the package source code, create a directory for patches
   `mkdir patches`
3. Import existing Buildroot patches
   `quilt import /path/to/buildroot/package/<foo>/*.patch`
4. Apply existing Buildroot patches
   `quilt push -a`
5. Create a new patch
   `quilt new 0001-fix-header-inclusion.patch`
6. Edit a file
   `quilt edit main.c`
7. Refresh the patch
   `quilt refresh`

User, permission and device tables

Package-specific users

- The default skeleton in `system/skeleton/` has a number of default users/groups.
- Packages can define their own custom users/groups using the `<pkg>_USERS` variable:

```
define <pkg>_USERS
        username uid group gid password home shell groups comment endef
```

- Examples:

```
define AVAHI_USERS
        avahi -1 avahi -1 * - - -
endef
```

```
define MYSQL_USERS
        mysql -1 nogroup -1 * /var/mysql - - MySQL daemon endef
```

File permissions and ownership

- By default, before creating the root filesystem images, Buildroot changes the ownership of all files to `0:0`, i.e. `root:root`
- Permissions are preserved as is, but since the build is executed as non-root, it is not possible to install setuid applications.
- A default set of permissions for certain files or directories is defined in `system/device_table.txt`.
- The `<pkg>_PERMISSIONS` variable allows packages to define special ownership and permissions for files and directories:

```
define <pkg>_PERMISSIONS
name type mode uid gid major minor start inc count endef
```

- The `major`, `minor`, `start`, `inc` and `count` fields are not used.

File permissions and ownership: examples

- `sudo` needs to be installed *setuid root*:

```
define SUDO_PERMISSIONS
        /usr/bin/sudo f 4755 0 0 - - - - -
endef
```

- `/var/lib/nginx` needs to be owned by `www-data`, which has UID/GID `33` defined in the skeleton:

```
define NGINX_PERMISSIONS
        /var/lib/nginx d 755 33 33 - - - - -
endef
```

Devices

- Defining devices only applies when the chosen `/dev` management strategy is *Static using a device table*. In other cases, *device files* are created dynamically.
- A default set of *device files* is described in `system/device_table_dev.txt` and created by Buildroot in the root filesystem images.
- When packages need some additional custom devices, they can use the `<pkg>_DEVICES` variable:

```
define <pkg>_DEVICES
name type mode uid gid major minor start inc count endef
```

- Becoming less useful, since most people are using a dynamic `/dev` nowadays.

Devices: example

xenomai.mk

```
define XENOMAI_DEVICES
/dev/rtheap  c  666  0  0  10  254  0  0  -
/dev/rtscope c  666  0  0  10  253  0  0  -
/dev/rtp     c  666  0  0  150 0    0  1  32
endef
```

Init scripts and systemd unit files

Init scripts, systemd unit files

- Buildroot supports several main init systems: *sysvinit*, *BusyBox*, *systemd*, *OpenRC*
- When packages want to install a program to be started at boot time, they need to install a startup script (*sysvinit*/*BusyBox*), a *systemd service* file, etc.
- They can do so using the following variables, which contain a list of shell commands.
  - `<pkg>_INSTALL_INIT_SYSV`
  - `<pkg>_INSTALL_INIT_SYSTEMD`
  - `<pkg>_INSTALL_INIT_OPENRC`
- Buildroot will execute the appropriate `<pkg>_INSTALL_INIT_xyz` commands of all enabled packages depending on the selected init system.

Init scripts, systemd unit files: example

bind.mk

```
define BIND_INSTALL_INIT_SYSV
	$(INSTALL) -m 0755 -D package/bind/S81named \
		$(TARGET_DIR)/etc/init.d/S81named
endef

define BIND_INSTALL_INIT_SYSTEMD
	$(INSTALL) -D -m 644 package/bind/named.service \
		$(TARGET_DIR)/usr/lib/systemd/system/named.service
endef
```

Config scripts

Config scripts: introduction

- Libraries not using `pkg-config` often install a **small shell script** that allows applications to query the compiler and linker flags to use the library.
- Examples: `curl-config`, `freetype-config`, etc.
- Such scripts will:
  - generally return results that are **not appropriate for cross-compilation**
  - be used by other cross-compiled Buildroot packages that use those libraries
- By listing such scripts in the `<pkg>_CONFIG_SCRIPTS` variable, Buildroot will **adapt the prefix, header and library paths** to make them suitable for cross-compilation.
- Paths in `<pkg>_CONFIG_SCRIPTS` are relative to `$(STAGING_DIR)/usr/bin`.

Config scripts: examples

libpng.mk

```
LIBPNG_CONFIG_SCRIPTS = \
        libpng$(LIBPNG_SERIES)-config libpng-config
```

imagemagick.mk

```
IMAGEMAGICK_CONFIG_SCRIPTS = \
        $(addsuffix -config,Magick MagickCore MagickWand Wand)

ifeq ($(BR2_INSTALL_LIBSTDCPP)$(BR2_USE_WCHAR),yy)
IMAGEMAGICK_CONFIG_SCRIPTS += Magick++-config
endif
```

Config scripts: effect

Without <pkg>_CONFIG_SCRIPTS

```
$ ./output/staging/usr/bin/libpng-config --cflags --ldflags
-I/usr/include/libpng16
-L/usr/lib -lpng16
```

With <pkg>_CONFIG_SCRIPTS

```
$ ./output/staging/usr/bin/libpng-config --cflags --ldflags
-I.../buildroot/output/host/arm-buildroot-linux-uclibcgnueabi/sysroot/usr/include/libpng16
-L.../buildroot/output/host/arm-buildroot-linux-uclibcgnueabi/sysroot/usr/lib -lpng16
```

Hooks

Hooks: principle (1)

- Buildroot *package infrastructure* often implement a default behavior for certain steps:
  - `generic-package` implements for all packages the download, extract and patch steps
  - Other infrastructures such as `autotools-package` or `cmake-package` also implement the configure, build and installations steps
- In some situations, the package may want to do **additional actions** before or after one of these steps.
- The **hook** mechanism allows packages to add such custom actions.

Hooks: principle (2)

- There are **pre** and **post** hooks available for all steps of the package compilation process:
  - download, extract, rsync, patch, configure, build, install, install staging, install target, install images, legal info
  - `<pkg>_(PRE|POST)_<step>_HOOKS`
  - Example: `CMAKE_POST_INSTALL_TARGET_HOOKS`, `CVS_POST_PATCH_HOOKS`, `BINUTILS_PRE_PATCH_HOOKS`
- Hook variables contain a list of make macros to call at the appropriate time.
  - Use `+=` to register an additional hook to a hook point
- Those make macros contain a list of commands to execute.

Hooks: examples

bind.mk: remove unneeded binaries

```
define BIND_TARGET_REMOVE_TOOLS
        rm -rf $(addprefix $(TARGET_DIR)/usr/bin/, $(BIND_TARGET_TOOLS_BIN))
endef

BIND_POST_INSTALL_TARGET_HOOKS += BIND_TARGET_REMOVE_TOOLS
```

vsftpd.mk: adjust configuration

```
define VSFTPD_ENABLE_SSL
        $(SED) 's/.*VSF_BUILD_SSL/#define VSF_BUILD_SSL/' \
                $(@D)/builddefs.h
endef

ifeq ($(BR2_PACKAGE_OPENSSL),y)
VSFTPD_DEPENDENCIES += openssl host-pkgconf VSFTPD_LIBS += `$(PKG_CONFIG_HOST_BINARY) --libs
libssl libcrypto`
VSFTPD_POST_CONFIGURE_HOOKS += VSFTPD_ENABLE_SSL
endif
```

Overriding commands

Overriding commands: principle

- In other situations, a package may want to completely **override** the default implementation of a step provided by a package infrastructure.
- A package infrastructure will in fact only implement a given step **if not already defined by a package**.
- So defining `<pkg>_EXTRACT_CMDS` or `<pkg>_BUILD_CMDS` in your package `.mk` file will override the package infrastructure implementation (if any).

Overriding commands: examples

jquery: source code is only one file

```
JQUERY_SITE = http://code.jquery.com JQUERY_SOURCE = jquery-$(JQUERY_VERSION).min.js

define JQUERY_EXTRACT_CMDS
        cp $(DL_DIR)/$(JQUERY_SOURCE) $(@D)
endef
```

tftpd: install only what's needed

```
define TFTPD_INSTALL_TARGET_CMDS
        $(INSTALL) -D $(@D)/tftp/tftp $(TARGET_DIR)/usr/bin/tftp
        $(INSTALL) -D $(@D)/tftpd/tftpd $(TARGET_DIR)/usr/sbin/tftpd
endef

$(eval $(autotools-package))
```

Legacy handling

Legacy handling: `Config.in.legacy`

- When a `Config.in` option is removed, the corresponding value in the `.config` is silently removed.
- Due to this, when users upgrade Buildroot, they generally don't know that an option they were using has been removed.
- Buildroot therefore adds the removed config option to `Config.in.legacy` with a description of what has happened.
- If any of these legacy options is enabled then Buildroot refuses to build.

DEVELOPERS file

DEVELOPERS file: principle

- A top-level `DEVELOPERS` file lists Buildroot developers and contributors interested in specific packages, board *defconfigs* or architectures.
- Used by:
  - The `utils/get-developers` script to identify to whom a patch on an existing package should be sent
  - The Buildroot *autobuilder* infrastructure to notify build failures to the appropriate package or architecture developers
- Important to add yourself in `DEVELOPERS` if you contribute a new package/board to Buildroot.

DEVELOPERS file: extract

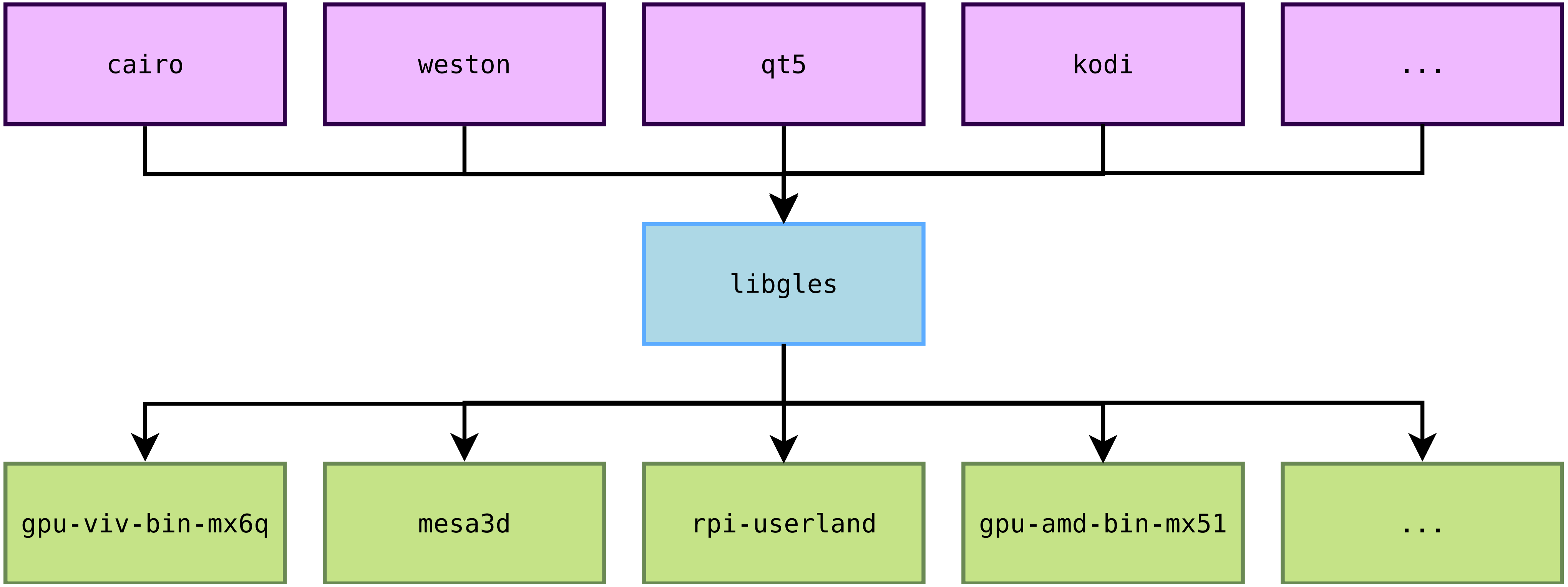
```
N:      Thomas Petazzoni <thomas.petazzoni@bootlin.com>
F:      arch/Config.in.arm F:      boot/boot-wrapper-aarch64/
F:      boot/grub2/
F:      package/android-tools/
F:      package/cmake/
F:      package/cramfs/
[...]
F:      toolchain/

N:      Waldemar Brodkorb <wbx@openadk.org>
F:      arch/Config.in.bfin F:      arch/Config.in.m68k F:      arch/Config.in.or1k F:
arch/Config.in.sparc F:      package/glibc/
```

Virtual packages

Virtual packages

- There are situations where different packages provide an implementation of the same interface
- The most useful example is OpenGL
  - OpenGL is an API
  - Each HW vendor typically provides its own OpenGL implementation, each packaged as separate Buildroot packages
- Packages using the OpenGL interface do not want to know which implementation they are using: they are simply using the OpenGL API
- The mechanism of *virtual packages* in Buildroot allows to solve this situation.
  - `libgles` is a virtual package offering the OpenGL ES API
  - Ten packages are *providers* of the OpenGL ES API: `gpu-amd-bin-mx51`, `imx-gpu-viv`, `gcnano-binaries`, `mali-t76x`, `mesa3d`, `nvidia-driver`, `rpi-userland`, `sunxi-mali-mainline`, `ti-gfx`, `ti-sgx-um`

Virtual packages



Virtual package definition: Config.in

libgles/Config.in

```
config BR2_PACKAGE_HAS_LIBGLES
        bool

config BR2_PACKAGE_PROVIDES_LIBGLES
        depends on BR2_PACKAGE_HAS_LIBGLES
        string
```

- `BR2_PACKAGE_HAS_LIBGLES` is a hidden boolean
  - Packages needing OpenGL ES will `depends on` it.
  - Packages providing OpenGL ES will `select` it.
- `BR2_PACKAGE_PROVIDES_LIBGLES` is a hidden string
  - Packages providing OpenGL ES will define their name as the variable value
  - The `libgles` package will have a build dependency on this provider package.

Virtual package definition: `.mk`

libgles/libgles.mk

```
$(eval $(virtual-package))
```

- Nothing to do: the `virtual-package` infrastructure takes care of everything, using the `BR2_PACKAGE_HAS_<name>` and `BR2_PACKAGE_PROVIDES_<name>` options.

Virtual package provider

sunxi-mali-mainline/Config.in

```
config BR2_PACKAGE_SUNXI_MALI_MAINLINE
        bool "sunxi-mali-mainline"
        select BR2_PACKAGE_HAS_LIBEGL
        select BR2_PACKAGE_HAS_LIBGLES

config BR2_PACKAGE_PROVIDES_LIBGLES
        default "sunxi-mali-mainline"
```

sunxi-mali-mainline/sunxi-mali-mainline.mk

```
[...]
SUNXI_MALI_MAINLINE_PROVIDES = libegl libgles
[...]
```

- The variable `<pkg>_PROVIDES` is only used to detect if two providers for the same virtual package are enabled.

Virtual package user

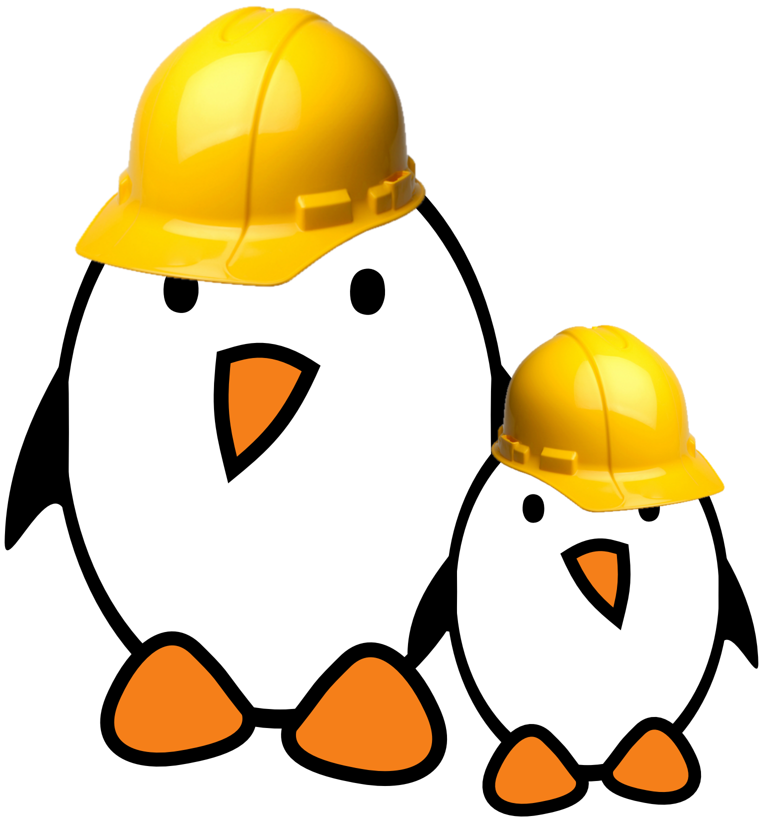
qt5/qt5base/Config.in

```
config BR2_PACKAGE_QT5BASE_OPENGL_ES2
        bool "OpenGL ES 2.0+"
        depends on BR2_PACKAGE_HAS_LIBGLES
        help
          Use OpenGL ES 2.0 and later versions.
```

qt5/qt5base/qt5base.mk

```
ifeq ($(BR2_PACKAGE_QT5BASE_OPENGL_DESKTOP),y)
QT5BASE_CONFIGURE_OPTS += -opengl desktop QT5BASE_DEPENDENCIES   += libgl
else ifeq ($(BR2_PACKAGE_QT5BASE_OPENGL_ES2),y)
QT5BASE_CONFIGURE_OPTS += -opengl es2
QT5BASE_DEPENDENCIES   += libgles
else QT5BASE_CONFIGURE_OPTS += -no-opengl
endif
```

Practical lab - Advanced packages



- Package an application with a mandatory dependency and an optional dependency
- Package a library, hosted on GitHub
- Use *hooks* to tweak packages
- Add a patch to a package